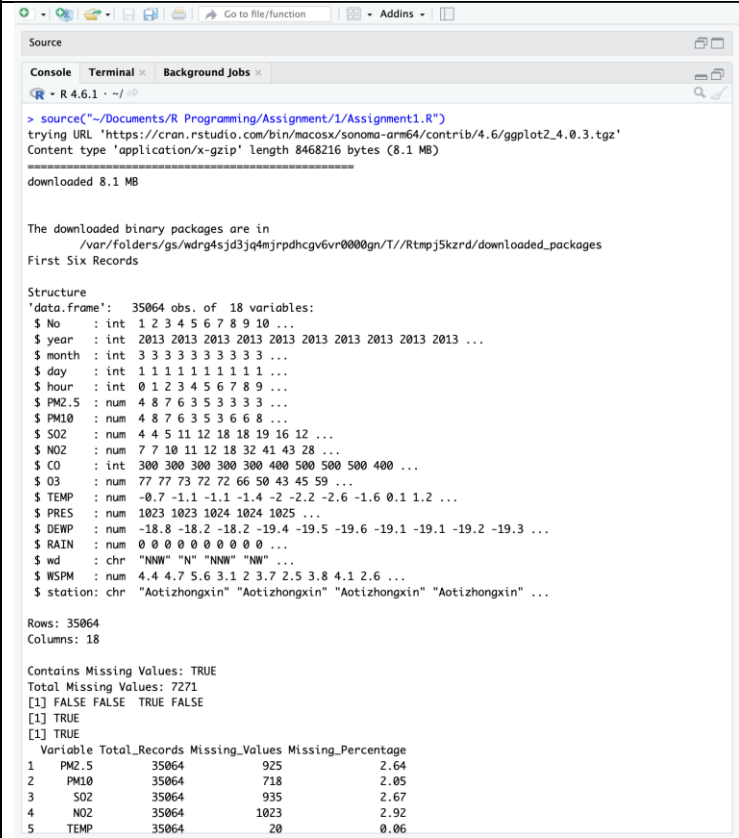


Semester	B.E. Semester VII - Computer Engineering
Subject	R Programming Lab
Subject In-charge	Dr Sanjeev Dwivedi

Student Name	SIDDHANTH KAUSHIK MUNGEKAR
Roll Number	23102B0030
Grade and Subject Teacher's Signature	

Assignment Number	01
Assignment Title	Air Quality Data Cleaning using R
Images	 <pre> > source("~/Documents/R Programming/Assignment/1/Assignment1.R") trying URL 'https://cran.rstudio.com/bin/macosx/sonoma-arm64/contrib/4.6/ggplot2_4.0.3.tgz' Content type 'application/x-gzip' length 8468216 bytes (8.1 MB) downloaded 8.1 MB The downloaded binary packages are in /var/folders/gs/wdrg4sjd3jq4mjrpdhcg6vr0000gn/T//Rtmpj5kzrd/downloaded_packages First Six Records Structure 'data.frame': 35064 obs. of 18 variables: \$ No : int 1 2 3 4 5 6 7 8 9 10 ... \$ year : int 2013 2013 2013 2013 2013 2013 2013 2013 2013 2013 ... \$ month : int 3 3 3 3 3 3 3 3 3 3 ... \$ day : int 1 1 1 1 1 1 1 1 1 1 ... \$ hour : int 0 1 2 3 4 5 6 7 8 9 ... \$ PM2.5 : num 4 8 7 6 3 5 3 3 3 3 ... \$ PM10 : num 4 8 7 6 3 5 3 6 6 8 ... \$ SO2 : num 4 4 5 11 12 18 18 19 16 12 ... \$ NO2 : num 7 7 10 11 12 18 32 41 43 28 ... \$ CO : int 300 300 300 300 300 400 500 500 500 400 ... \$ O3 : num 77 77 73 72 72 66 50 43 45 59 ... \$ TEMP : num -0.7 -1.1 -1.1 -1.4 -2 -2.2 -2.6 -1.6 0.1 1.2 ... \$ PRES : num 1023 1023 1024 1024 1025 ... \$ DEWP : num -18.8 -18.2 -18.2 -19.4 -19.5 -19.6 -19.1 -19.2 -19.3 ... \$ RAIN : num 0 0 0 0 0 0 0 0 0 ... \$ wd : chr "NNW" "N" "NNW" "NW" ... \$ WSPM : num 4.4 4.7 5.6 3.1 2 3.7 2.5 3.8 4.1 2.6 ... \$ station : chr "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" ... Rows: 35064 Columns: 18 Contains Missing Values: TRUE Total Missing Values: 7271 [1] FALSE FALSE TRUE FALSE [1] TRUE [1] TRUE Variable Total_Records Missing_Values Missing_Percentage 1 PM2.5 35064 925 2.64 2 PM10 35064 718 2.05 3 SO2 35064 935 2.67 4 NO2 35064 1023 2.92 5 TEMP 35064 20 0.06 </pre>

```
Source
Console Terminal Background Jobs
R - R 4.6.1 - ~/
Variable Total_Records Missing_Values Missing_Percentage
1 PM2.5 35064 925 2.64
2 PM10 35064 718 2.05
3 SO2 35064 935 2.67
4 NO2 35064 1023 2.92
5 TEMP 35064 20 0.06
6 WSPM 35064 14 0.04
7 wd 35064 81 0.23
NA : 934
NaN : 0
Infinite : 0

Variable : PM2.5
Missing Before : 925
Median : 58
Missing After : 0

Variable : PM10
Missing Before : 718
Median : 87
Missing After : 0

Variable : SO2
Missing Before : 935
Median : 9
Missing After : 0

Variable : NO2
Missing Before : 1023
Median : 53
Missing After : 0

Variable : TEMP
Missing Before : 20
Median : 14.5
Missing After : 0

Variable : WSPM
Missing Before : 14
Median : 1.4
Missing After : 0

Wind Direction Mode : NE
Before : 81
After : 0
Variable Missing_Before Missing_After Values_Replaced

Wind Direction Mode : NE
Before : 81
After : 0
Variable Missing_Before Missing_After Values_Replaced
1 PM2.5 925 0 925
2 PM10 718 0 718
3 SO2 935 0 935
4 NO2 1023 0 1023
5 TEMP 20 0 20
6 WSPM 14 0 14
7 wd 81 0 81

Dataset exported successfully.
>
```

Collab Solution:

```
... Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

First Six Records

A data.frame: 6 x 18

	No	year	month	day	hour	PM2.5	PM10	S02	N02	CO	O3	TEMP	PRES	DEWP	RAIN	wd	WSPM	station
	<int>	<int>	<int>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<chr>
1	1	2013	3	1	0	4	4	4	7	300	77	-0.7	1023.0	-18.8	0	NNW	4.4	Aotizhongxin
2	2	2013	3	1	1	8	8	4	7	300	77	-1.1	1023.2	-18.2	0	N	4.7	Aotizhongxin
3	3	2013	3	1	2	7	7	5	10	300	73	-1.1	1023.5	-18.2	0	NNW	5.6	Aotizhongxin
4	4	2013	3	1	3	6	6	11	11	300	72	-1.4	1024.5	-19.4	0	NW	3.1	Aotizhongxin
5	5	2013	3	1	4	3	3	12	12	300	72	-2.0	1025.2	-19.5	0	N	2.0	Aotizhongxin
6	6	2013	3	1	5	5	5	18	18	400	66	-2.2	1025.6	-19.6	0	N	3.7	Aotizhongxin

```
... Structure
```

```
'data.frame': 35064 obs. of 18 variables:
```

```
$ No      : int  1 2 3 4 5 6 7 8 9 10 ...
$ year    : int 2013 2013 2013 2013 2013 2013 2013 2013 2013 2013 ...
$ month   : int  3 3 3 3 3 3 3 3 3 3 ...
$ day     : int  1 1 1 1 1 1 1 1 1 1 ...
$ hour    : int  0 1 2 3 4 5 6 7 8 9 ...
$ PM2.5   : num  4 8 7 6 3 5 3 3 3 3 ...
$ PM10    : num  4 8 7 6 3 5 3 6 6 8 ...
$ S02     : num  4 4 5 11 12 18 18 19 16 12 ...
$ N02     : num  7 7 10 11 12 18 32 41 43 28 ...
$ CO      : int 300 300 300 300 300 300 400 500 500 500 ...
$ O3      : num  77 77 73 72 72 66 50 43 45 59 ...
$ TEMP    : num -0.7 -1.1 -1.1 -1.4 -2 -2.2 -2.6 -1.6 0.1 1.2 ...
$ PRES    : num 1023 1023 1024 1024 1025 ...
$ DEWP    : num -18.8 -18.2 -18.2 -19.4 -19.5 -19.6 -19.1 -19.1 -19.2 -19.3 ...
$ RAIN    : num  0 0 0 0 0 0 0 0 0 0 ...
$ wd      : chr  "NNW" "N" "NNW" "NW" ...
$ WSPM    : num  4.4 4.7 5.6 3.1 2 3.7 2.5 3.8 4.1 2.6 ...
$ station : chr  "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" "Aotizhongxin" ...
```

```
Rows: 35064
```

```
Columns: 18
```

```
... Contains Missing Values: TRUE
Total Missing Values: 7271
[1] FALSE FALSE TRUE FALSE
[1] TRUE
[1] TRUE
Variable Total_Records Missing_Values Missing_Percentage
1 PM2.5 35064 925 2.64
2 PM10 35064 718 2.05
3 S02 35064 935 2.67
4 N02 35064 1023 2.92
5 TEMP 35064 20 0.06
6 WSPM 35064 14 0.04
7 wd 35064 81 0.23
NA : 934
NaN : 0
Infinite : 0
```

```
... Variable : PM2.5
Missing Before : 925
Median : 58
Missing After : 0

Variable : PM10
Missing Before : 718
Median : 87
Missing After : 0

Variable : S02
Missing Before : 935
Median : 9
Missing After : 0

Variable : N02
Missing Before : 1023
Median : 53
Missing After : 0

Variable : TEMP
Missing Before : 20
Median : 14.5
Missing After : 0

Variable : WSPM
Missing Before : 14
Median : 1.4
Missing After : 0
```

```

Wind Direction Mode : NE
Before : 81
After : 0
4 8 7 6 3 5 3 3 3 3 3 3 3 6 8 9 10 11 8 11 12 15 24 22 14 13 3 3 9 4 3 3 10 11 18 26 25 26 37 37 44 54 61 67 74 81 93 112 109 110 105
106 101 91 79 77 83 94 96 91 99 72 48 28 26 24 19 60 117 54 46 42 49 34 12 7 10 13 14 13 15 16 18 13 3 3 10 14 20 25 31 34 40 51 58 71 91
100 103 115 117 104 106 111 114 131 138 142 163 168 165 166 190 203 210 205 215 219 226 284 272 242 212 192 188 186 188 194 184 175 179 192 201
230 248 248 230 239 242 254 266 254 260 250 240 238 243 246 275 306 320 292 251 228 217 255 258 277 282 267 216 201 211 227 282 363 376 344 339
339 324 322 315 310 309 310 326 326 219 104 121 184 330 172 114 82 64 91 61 90 79 49 89 89 96 97 102 110 108 ... 58 10 58 15 16 20 25 28 58 58
46 50 58 58 90 58 58 74 102 58 91 58 42 40 28 21 29 31 47 67 60 62 55 70 81 88 86 86 91 95 95 108 114 131 142 139 147 145 131 28 11 10 14 14
15 12 13 15 11 12 15 13 13 11 15 12 11 13 13 12 3 18 31 23 10 6 33 26 30 45 46 48 36 36 28 23 16 20 20 26 23 22 21 16 16 18 11 11 9 11 10 14
18 18 12 7 11 7 13 13 3 11 5 7 17 9 12 7 10 8 9 11 14 14 17 12 25 15 24 22 19 18 18 25 30 32 30 19 20 31 26 23 24 24 26 29 30 30 36 42 51
46 38 55 77 85 94 96 80 73 61 52 69 122 58 58 103 95 75 58 47 51 51 53 58 70 56 15 9 7 13 12 9 11 13 12 11 7 6 6 5 6 12 12 13 12 13 16 21 19
Variable Missing_Before Missing_After Values_Replaced
1 PM2.5 925 0 925
2 PM10 718 0 718
3 SO2 935 0 935
4 NO2 1823 0 1823
5 TEMP 28 0 28
6 WSPH 14 0 14
7 wd 81 0 81

```

Dataset exported successfully.

Missing Values Before and After Cleaning

